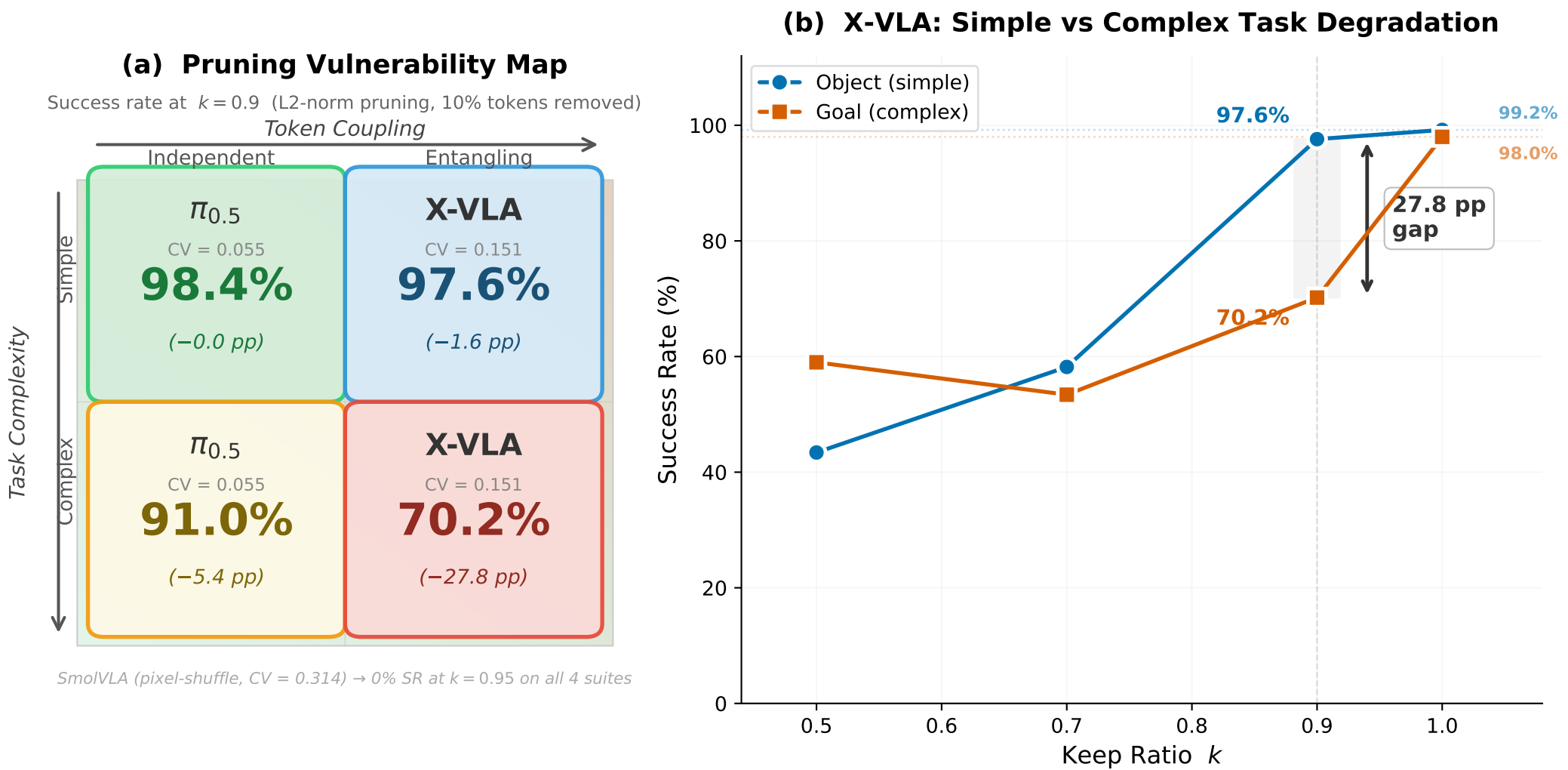

(a) Pruning Vulnerability Map
Success rate at $k = 0.9$ (L2-norm pruning, 10% tokens removed)
Token Coupling
Independent
Entangling
Task Complexity
Simple
Complex
$\pi_{0.5}$
CV = 0.055
98.4%
(−0.0 pp)
X-VLA
CV = 0.151
97.6%
(−1.6 pp)
$\pi_{0.5}$
CV = 0.055
91.0%
(−5.4 pp)
X-VLA
CV = 0.151
70.2%
(−27.8 pp)
SmolVLA (pixel-shuffle, CV = 0.314) → 0% SR at $k = 0.95$ on all 4 suites
(b) X-VLA: Simple vs Complex Task Degradation
Object (simple)
Goal (complex)
97.6%
99.2%
98.0%
70.2%
27.8 pp gap
Success Rate (%)
Keep Ratio k
0
20
40
60
80
100
0.5
0.6
0.7
0.8
0.9
1.0